



Polypharmacy (herbal and synthetic drug combination): a novel approach in the treatment of type-2 diabetes and its complications in rats

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Abstract The aim of present study was to evaluate the effect of aqueous leaf extract of *Annona squamosa* with Glipizide in a high fat diet and streptozocin-induced type-2 diabetes. Nine groups ($n = 6$) of male Sprague-Dawley rats were used for the study, with. Basal blood glucose, urine volume, and body weights were measured and the rate were kept on a high fat diet. After 15 days, streptozocin in sub-diabetic dose (35 mg/kg) was administered to the animals to induce diabetes. With 1 week of consistent hyperglycemia, treatment was initiated. Aqueous extract of *Annona squamosa* was administered orally at 350 mg/kg body weight alone and in combination with reduced and reducing dose combinations of Glipizide. Blood glucose, body weight, urine volume were measured every 10th day. The elevated blood-glucose level in diabetic rats was controlled better with combination therapy compared with

the synthetic drug alone or the herbal stand-alone drug. All the results were statistically significant ($P < 0.001$). A combination of *Annona squamosa* along with Glipizide may be helpful in dose reduction of Glipizide up to 50 %, reducing the risk of the onset of insulin therapy.

Keywords *Annona squamosa* · Glipizide · Combination therapy · Biochemical parameter

Introduction

Diabetes mellitus is metabolic disorder of fat, carbohydrate, and protein that results from defects in insulin secretion, insulin action (sensitivity), or both [1]. Different types of oral hypoglycemic agents, such as biguanides, sulfonylurea, glucosidase inhibitors, thiazolidinedione derivatives, etc., are available along with insulin for the treatment of diabetes mellitus. The use of any pharmacological therapy for type 2 diabetes appears to be associated with an increased risk of heart or kidney failure. Long-term treatment by insulin therapy is associated with several drawbacks, such as insulin resistance, brain atrophy, anorexia, fatty liver, etc. [2].

Herbal drugs can have beneficial effects in terms of reducing the dosage, side effects and duration of action of synthetic drugs [3]. There are various studies in which a combination of herbal drugs and synthetic drugs is used to treat a disorder, and researchers have found better results compared to the synthetic drug alone. Herbal drugs can elevate the bioavailability of a synthetic drug [4]. Combination drugs can decrease the chances of pancreatitis and thereby are helpful in insulin protection. Curcumin administered in combination with Rosiglitazone showed better results in induced acute pancreatitis [5].

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Furthermore, herbal drugs could have some additive effects in the form of synergism of synthetic drugs. Curcumin potentiates the analgesic activity of Diclofenac in a formalin model of facial pain in rats [6]. Curcumin and β -carotene shows better effects on the pharmacodynamics and pharmacokinetics of Phenytoin when given in combination in maximal electroshock seizures in rats [7]. Herbal drugs can directly decrease absorption of glucose from the intestine [8]. Herbal and synthetic drug combinations can increase or decrease the therapeutic efficacy of each other [9]. Propranolol and hydrochlorothiazide in combination with garlic homogenate decreases the hypolipidemic action of garlic homogenate, while administration of garlic homogenate along with captopril augments its hypolipidemic effect in rats [10]. Further, captopril with garlic shows a cardioprotective interaction in isoproterenol-induced myocardial infarction in rats, and the combination dislodges the effect of isoproterenol on superoxide dismutase (SOD) and catalase (CAT) and retained the activities of lactate dehydrogenase (LDH) and creatine kinase isoenzyme MB (CK-MB) [11]. The World Health Organisation (WHO) approves the use of herbal drugs for treatment of different diseases including diabetes [12].

Annona squamosa (custard apple) is a native of West Indies and is cultivated throughout India, mainly for its edible fruit. This plant possesses several medicinal properties [13]. A couple of flavonoids [14] and a tetrahydro-isoquinoline alkaloid with cardiogenic activity [15] have been isolated from the leaves of *A. squamosa*, which is also used as an insecticidal agent [16]. Partially purified flavonoids of aqueous *A. squamosa* leaves extract possess antimicrobial and insecticidal potential [17]. Quercetin is antidiabetic principle present in *A. squamosa* [18]. Acetogenins are potent pesticides, antimalarial, parasiticidal, anti-microbial, cytotoxic and immunosuppressive activities [19–21]. Diterpenes isolated from the plant have anti-HIV principle and anti-platelet aggregation potential [22, 23].

Combination studies are helpful to establish the safety and cost effectiveness of drug therapies. In the present study, we performed experiments on high-fat-diet diabetic rats using a subdiabetic dose of streptozocin and aqueous extracts of *A. squamosa*, to examine the beneficial effects of this new combination therapy in diabetic complications.

Materials and methods

Chemicals

Cholesterol was obtained from SD Fine Chemicals (Mumbai, India), DL methionine Himedia Lab (Mumbai,

India), vitamins, minerals and coconut oil were purchased from a local market, and streptozotocin from MP Biochem (USA). Glipizide and Metformin were provided as a gift sample from Zydus Research Centre, Ahmedabad, India. Kits for measuring the biochemical parameters were purchased (Nicholas India, Basheerbagh, Hyderabad, India). All chemicals were of analytical grade.

Preparation of extract

Annona squamosa leaves were collected from a local market in Dehradun and were authenticated by Dr. Imran Kazmi (Assistant Professor), Siddhartha Institute of Pharmacy, Dehradun, India. A voucher specimen (SIP/DOEP/consult/20-02-11/1197/297) was deposited in the herbarium of Siddhartha Institute of Pharmacy, Dehradun, India. An aqueous extract was prepared by the cold maceration method. Shade-dried leaf powder (250 g) macerated in 750 ml distilled water was allowed to stand overnight, then boiled for 5–10 min until the volume was reduced to half its original volume, cooled, filtered, concentrated, dried in the vacuum (yield 60 g), and the extract was stored at 2–8 °C.

Animals

Male Sprague-Dawley rats 150–200 g in weight were procured from the Animal House of the Siddhartha Institute of Pharmacy, Dehradun. The animals were maintained in controlled temperature as well as humidity and with standard diet and water ad libitum. The protocol was approved by the Institutional Animal Ethical Committee (IAEC) as per the guidance of Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA); Ministry of Social Justice and Empowerment, Government of India. All animals were age-matched and divided into nine groups ($n = 6$). Group I contained normal animals and served as a normal control (NC). Group II served as a diabetic control (DC). Groups I and II received vehicle during the experiments, while Group III received the reference standard drug Glipizide (5 mg/kg), Group IV received aqueous extracts of *A. squamosa* (350 mg/kg), Group V (3.75 mg/kg Glipizide + *A. squamosa*) received 3.75 mg/kg dose of Glipizide and *A. squamosa*, Group VI (2.5 mg/kg Glipizide + *A. squamosa*) received 2.5 mg/kg dose of Glipizide + *A. squamosa*, Group VII received 1.25 mg/kg dose of Glipizide + *A. squamosa*, Group VIII Glipizide (5 mg/kg) and Metformin (50 mg/kg), and Group IX (reducing dose of Glipizide + *A. squamosa*) received *A. squamosa* (0.0 mg/kg, 75 mg/kg, 150 mg/kg, 225 mg/kg) and Glipizide (5 mg/kg, 3.75 mg/kg, 2.5 mg/kg, 1.25 mg/kg) every 10th day. The protocol was set for 30 days.

Extract standardisation by high performance liquid chromatography

Preparation of standard mix (rutin and quercetin) and sample solution

Working standard solutions of quercetin and rutin were prepared by dissolving 5 mg standard rutin and quercetin in 5 ml high performance liquid chromatography (HPLC) grade methanol and sonicating. The final volume was adjusted with methanol. Sample solutions were prepared by extracting 2.5 g *A. squamosa* leaves in 50 ml water and refluxing for 1 h at 70 °C. The extract was filtered through Whatman filter paper and evaporated to dryness in a water bath. Dried extract was transferred to a 10-ml volumetric flask, about 5 ml HPLC grade methanol was added, sonicated and final volume was adjusted with methanol.

Procedure

Equal volumes (10 µl) of blank (methanol), standard and sample solution were injected into the chromatograph and the chromatogram recorded. The peak in the sample chromatogram at the same retention time as that observed in the blank chromatogram was disregarded.

Induction of diabetes in rats

Healthy Sprague-Dawley rats showing normal plasma glucose level in the range of 80–120 mg/dl were selected. They were fed a high fat diet (58 % fat derived calories) for 2 weeks prior to streptozotocin injection. A single dose of streptozotocin (35 mg/kg, i.p.) was administered for induction of diabetes. Blood glucose levels were measured after 48 h of streptozotocin administration. Animals showing fasting blood glucose more than or equal to 200 mg/dl were considered as diabetic and were used for studies. The high fat diet continued throughout the experiment.

Estimation of biochemical parameters

Blood samples were collected upon termination of the experiment from retro-orbital plexus under light ether anesthesia without any anticoagulant and was allowed to stand for 30 min at room temperature, centrifuged at 2,500 rpm for 10 min to separate the serum. The serum obtained was kept at 2–4 °C for further use. Blood glucose level was measured using digital glucometer (Chopra Pharmaceuticals, New Delhi, India). A drop of blood was placed on one side of the test strip to be inserted in the glucometer. The glucose level was displayed on the screen within 20 s. Serum total cholesterol, triglyceride, and high density lipoprotein (HDL)-cholesterol were estimated using a standard kit (Nicholas

India) with a semi-auto analyser (photometer 5010, Nicholas India Pvt). Antioxidant, serum glutamate pyruvate transaminase (SGPT), serum glutamate oxaloacetate transaminase (SGOT) parameters were also measured. Along with these parameters behavioural studies were conducted.

Statistical analysis

Results were expressed as mean $\pm$ SEM. Statistical significance between more than two groups was tested using one-way ANOVA followed by the Bonferroni multiple comparisons test or unpaired two-tailed student's *t* test as appropriate using computer based fitting program (Prism, Graphpad, <http://www.graphpad.com>). Differences were considered to be statistically significant when $P < 0.05$.

Results

Quantification of rutin and quercetin in aqueous extract of leaves of *Annona squamosa* by HPLC.

Peaks for rutin and quercetin were determined in aqueous extract of *A. squamosa* and identified by comparing standard peaks of rutin and quercetin (Fig. 1).

Blood glucose levels

Blood glucose levels are shown in Table 1. A significant increase in the levels of blood glucose was observed in diabetic rats when compared to control rats ($P < 0.001$). Administration of *A. squamosa* alone and with reduced (3.75 mg/kg, 2.5 mg/kg, 1.25 mg/kg) and reducing (0.0 mg/kg, 1.25 mg/kg, 2.5 mg/kg, 3.75 mg/kg) dose combination of Glipizide in diabetic rats decreased significantly the level of blood glucose ($P < 0.001$). Blood glucose level control was nearly the same with the reduced dose combination of Glipizide with *A. squamosa* as with Glipizide alone. In the reducing dose combination, blood glucose was also very similar to animals treated with Glipizide alone. No significant alteration in the body weights of the animals was found.

Kidney parameters

Lipid peroxidation

Table 2 shows the concentration of lipid peroxidation in kidneys of both control and experimental groups of rats. There was a significant ($P < 0.001$) elevation in tissue lipid peroxidation in diabetic rats. Administration of *A. squamosa* alone reduced and reducing dose of Glipizide in combination with *A. squamosa* significantly ($P < 0.01$) reduced the elevated levels of lipid peroxidase in diabetic rats. Reduced dose of Glipizide 3.75 mg/kg and 2.5 mg/kg

Fig. 1 High performance liquid chromatography profile of aqueous extract of the leaves of *Annona squamosa* (custard apple)

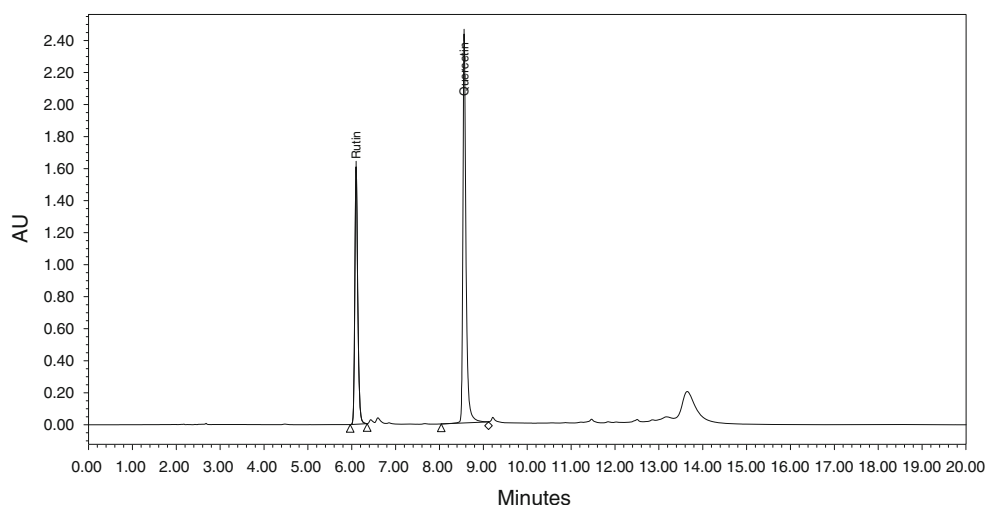


Table 1 Blood glucose levels at different dose combinations of *Annona squamosa* and Glipizide

Group	Blood glucose level (mg/dl)	
	Initial	Final
Normal control	093.64 ± 8.71	089.12 ± 2.51
Diabetic control	305.01 ± 7.11****	323.68 ± 2.10****
5 mg/kg GLZ	321.11 ± 3.21*	144.74 ± 3.69***
300 mg/kg AS	348.20 ± 2.85**	212.41 ± 2.48***
3.75 mg/kg GLZ + 300 mg/kg AS	299.01 ± 1.40*	210.22 ± 3.10***
2.5 mg/kg GLZ + 300 mg/kg AS	278.30 ± 2.39**	159.75 ± 0.73***
1.25 mg/kg GLZ + 300 mg/kg AS	219.21 ± 1.61***	189.66 ± 2.86***
3.75 mg/kg GLZ + 50 mg/kg MET	227.32 ± 2.6***	121.30 ± 1.78***
3.75/2.5/1.25 mg/kg GLZ + 300 mg/kg AS	241.41 ± 3.63***	177.71 ± 3.51***

Each value represents the mean ± SEM for each group ($n = 6$)

GLZ Glipizide, MET metformin, AS *Annona squamosa*, 3.75/2.5/1.25 mg/kg GLZ: reduction in dose of Glipizide every 10th day

* $P < 0.05$ diabetic control vs treated group, ** $P < 0.01$ diabetic control vs treated group, *** $P < 0.001$ diabetic control vs treated group, **** $P < 0.001$ normal vs diabetic control

in combination with *A. squamosa* showed similar results compared to Glipizide alone. The results of reducing dose combination of Glipizide with *A. squamosa* were very similar to Glipizide alone (see Fig. 2).

Super oxide dismutase

The concentration of SOD in tissues decreased significantly ($P < 0.001$, $P < 0.01$) in diabetic rats when compared with control group. Administration of *A. squamosa* alone and in reduced and reducing dose combination of Glipizide with *A. squamosa* extract showed better results as compared to Glipizide alone (Table 2).

Catalase and glutathione

A significant ($P < 0.001$) lowering in the levels of CAT and glutathione (GSH) was observed in diabetic animals

compare to normal controls. Treatment with *A. squamosa* extract alone, with reduced and reducing dose combinations of Glipizide significantly ($P < 0.001$, $P < 0.01$) increased the concentration of these enzymes. In reduced dose of Glipizide 3.75 mg/kg and 2.5 mg/kg and reducing doses with *A. squamosa* showed better elevated levels of CAT compared to Glipizide alone, while GSH levels were also controlled better in 3.75 mg/kg and 2.5 mg/kg dose combinations of Glipizide with *A. squamosa*. The results of the reducing dose combination were very similar to those of Glipizide alone (Table 2).

Liver parameters

Table 3 and Fig. 3 show the levels of SGOT and SGPT in serum of normal control, streptozotocin diabetic rats, and other treatment groups. The activities of these enzymes were found to be increased significantly ($P < 0.001$) in the

Table 2 Antioxidant enzymes levels of different dose combination of *A. squamosa* and Glipizide

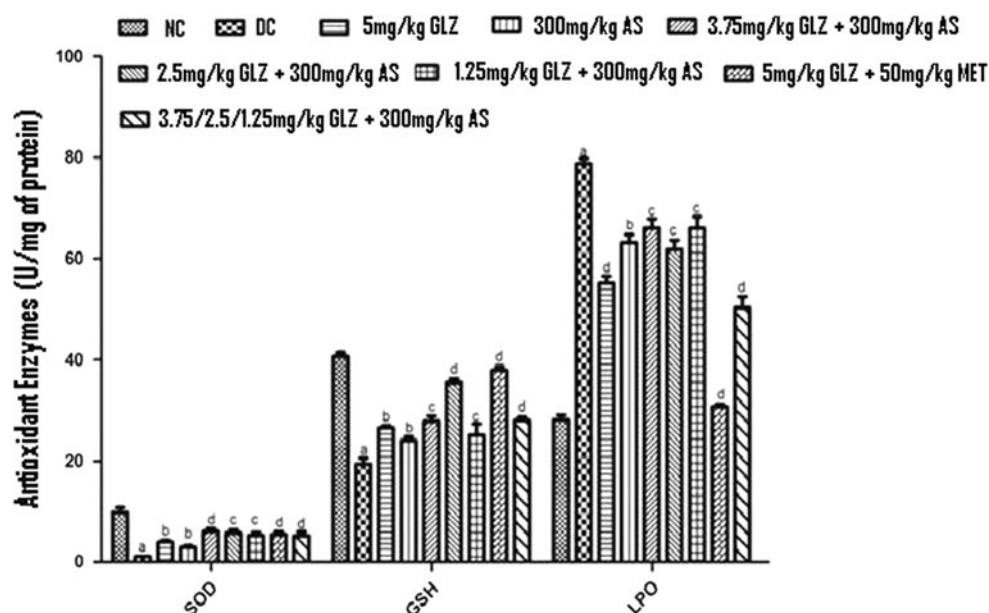
Group	Antioxidant enzymes			
	SOD (μ mg protein ⁻¹)	CAT (nm min ⁻¹ mg protein ⁻¹)	GSH (μ mg protein ⁻¹)	LPO (nm mg protein ⁻¹)
Normal control	9.79 $\pm$ 1.01	0.010 $\pm$ 0.0004	40.70 $\pm$ 0.78	28.13 $\pm$ 0.91
Diabetic control	1.12 $\pm$ 0.05****	0.002 $\pm$ 0.0010****	19.34 $\pm$ 1.21****	75.78 $\pm$ 1.06****
5 mg/kg GLZ	3.94 $\pm$ 0.38*	0.011 $\pm$ 0.0005***	26.55 $\pm$ 0.48*	55.16 $\pm$ 1.37***
300 mg/kg AS	2.91 $\pm$ 0.38*	0.009 $\pm$ 0.0004***	24.04 $\pm$ 0.80*	63.22 $\pm$ 1.52*
3.75 mg/kg GLZ + 300 mg/kg AS	6.12 $\pm$ 0.57***	0.010 $\pm$ 0.0002***	27.88 $\pm$ 1.01**	66.06 $\pm$ 1.81**
2.5 mg/kg GLZ + 300 mg/kg AS	5.79 $\pm$ 0.59**	0.010 $\pm$ 0.0007***	35.53 $\pm$ 0.72***	61.85 $\pm$ 1.80**
1.25 mg/kg GLZ + 300 mg/kg AS	5.27 $\pm$ 0.62**	0.009 $\pm$ 0.0004***	25.05 $\pm$ 2.28**	66.05 $\pm$ 2.30**
3.75 mg/kg GLZ + 50 mg/kg MET	5.46 $\pm$ 0.60***	0.014 $\pm$ 0.0007***	37.95 $\pm$ 0.79***	30.75 $\pm$ 0.49***
3.75/2.5/1.25 mg/kg GLZ + 300 mg/kg AS	5.24 $\pm$ 0.78***	0.009 $\pm$ 0.0001***	28.04 $\pm$ 0.69***	50.24 $\pm$ 2.31***

Each value represents the mean $\pm$ SEM for each group ($n = 6$)

GLZ Glipizide, MET metformin, AS *Annona squamosa*, SOD superoxide dismutase, CAT catalase, GSH glutathione, LPO lipid peroxidation, 3.75/2.5/1.25 mg/kg GLZ: reduction in dose of Glipizide every 10th day

* $P < 0.05$ diabetic control vs treated group, ** $P < 0.01$ diabetic control vs treated group, *** $P < 0.001$ diabetic control vs treated group, **** $P < 0.001$ normal vs diabetic control

Fig. 2 Antioxidant enzymes levels at different dose combinations of *A. squamosa* and glipizide. NC Normal control, DC diabetic control, GLZ Glipizide, MET Metformin, ASA *squamosa*, SOD superoxide dismutase, GSH Glutathione, LPO lipid peroxidation, 3.75/2.5/1.25 mg/kg GLZ: reduction in dose of Glipizide every 10th day. *a* $P < 0.001$ normal vs diabetic control, *b* $P < 0.05$ diabetic control vs treated group, *c* $P < 0.01$ diabetic control vs treated group, *d* $P < 0.001$ diabetic control vs treated group



serum of diabetic rats, when compared with other groups. There were no significant changes in the total bilirubin levels of the animals, which were all in the normal range (Table 3).

Serum glutamate pyruvate transaminase

The elevated levels of SGPT after oral administration of *A. squamosa* alone and in combination with Glipizide for 30 days returned significantly ($P < 0.001$, $P < 0.01$) near to normal. The reduced dose combination of 3.75 mg/kg

Glipizide with *A. squamosa* controlled the SGPT levels similarly to Glipizide alone.

Serum glutamate oxaloacetate transaminase

SGOT levels in diabetic rats were elevated ($P < 0.001$) compared to normal controls. Oral administration of *A. squamosa* alone, in reduced and reducing dose combination of Glipizide with *A. squamosa* controlled elevated SGOT levels. Better activity was found in the reduced and reducing dose combination of Glipizide with *A. squamosa* as compared to Glipizide alone.

Table 3 Levels of liver marker enzymes and total bilirubin at different dose combination of *Annona squamosa* and glipizide

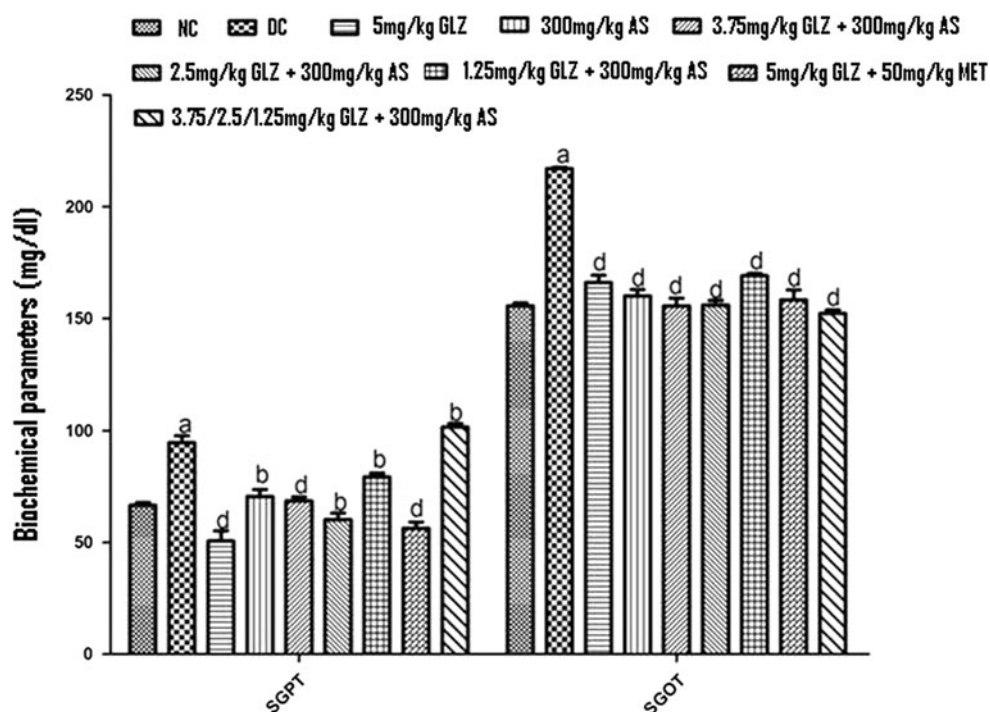
Group	Biochemical parameters		
	SGPT (mg/dl)	SGOT (mg/dl)	T BILI (mg/dl)
Normal control	066.66 ± 1.18	155.59 ± 1.20	0.16 ± 0.03
Diabetic control	094.57 ± 3.08****	217.11 ± 0.57****	0.13 ± 0.03 ns
5 mg/kg GLZ	050.64 ± 4.60***	166.23 ± 3.20***	0.20 ± 0.05 ns
300 mg/kg AS	070.61 ± 3.11*	160.18 ± 2.83***	0.21 ± 0.01 ns
3.75 mg/kg GLZ + 300 mg/kg AS	068.56 ± 1.77***	155.60 ± 3.44***	0.22 ± 0.05 ns
2.5 mg/kg GLZ + 300 mg/kg AS	060.30 ± 2.83*	156.16 ± 2.00***	0.16 ± 0.03 ns
1.25 mg/kg GLZ + 300 mg/kg AS	079.33 ± 1.84*	169.18 ± 1.10***	0.20 ± 0.00 ns
3.75 mg/kg GLZ + 50 mg/kg MET	056.32 ± 2.91***	158.13 ± 4.40***	0.13 ± 0.03 ns
3.75/2.5/1.25 mg/kg GLZ + 300 mg/kg AS	101.63 ± 1.60 *	152.30 ± 1.40***	0.23 ± 0.03 ns

Each value represents the mean ± SEM for each group ($n = 6$)

SGPT Serum glutamate pyruvate transaminase, SGOT serum glutamate oxaloacetate transaminase, T BILI total bilirubin, GLZ Glipizide, MET Metformin, AS *Annona squamosa*, 3.75/2.5/1.25 mg/kg GLZ: reduction in dose of Glipizide every 10th day

* $P < 0.05$ diabetic control vs treated group, ** $P < 0.01$ diabetic control vs treated group, *** $P < 0.001$ diabetic control vs treated group, **** $P < 0.001$ normal vs diabetic control

Fig. 3 Liver marker enzymes levels at different dose combination of *Annona squamosa* and glipizide, NC normal control, DC diabetic control, GLZ Glipizide, MET Metformin, AS *Annona squamosa*, SGPT serum glutamate pyruvate transaminase, SGOT serum glutamate oxaloacetate transaminase, 3.75/2.5/1.25 mg/kg GLZ: reduction in dose of Glipizide every 10th day. *a* $P < 0.001$ normal versus diabetic control, *b* $P < 0.05$ diabetic control versus treated group, *c* $P < 0.01$ diabetic control versus treated group, *d* $P < 0.001$ diabetic control versus treated group



Behavioural (psychological) study

Elevated plus maze test for anxiety

Anxiety of animals was decreased significantly as assessed by open arm exploration time in the diabetic control rats ($P < 0.001$) (Table 4). Oral administration of *A.squamosa* alone and reduced and reducing dose combination of Glipizide with *A. squamosa* for 30 days improved the anxiety behaviour of the diabetic rats. In the elevated plus

maze study after 15 days of the treatment the *A. squamosa* alone treated group showed better activity among all groups. The results of reduced and reducing dose combinations were nearly similar to the Glipizide alone. After 30 days treatment the results of reduced dose combination 3.75 mg/kg and 2.5 mg/kg of Glipizide with *A. squamosa* were better as compare to the Glipizide alone and *A. squamosa* alone. The reducing dose combination of Glipizide with *A. squamosa* showed the nearly similar results as compare to Glipizide alone.

Table 4 Behavioural parameters at different dose combination of *A. squamosa* and glipizide

Group	Behavioural (psychological) parameters			
	Elevated plus maze test		Forced swim test	
	Open arms exploration time (s)		Time of immobilization (s)	
	15th day of treatment	30th day of treatment	15th day of treatment	30th day of treatment
Normal control	65.00 ± 7.58	76.33 ± 6.91	442.68 ± 12.11	419.73 ± 16.91
Diabetic control	14.33 ± 7.42****	3.33 ± 1.70****	216.71 ± 18.48****	149.30 ± 14.7****
5 mg/kg GLZ	52.00 ± 1.50**	52.01 ± 3.58***	306.70 ± 14.82**	278.72 ± 49.77**
300 mg/kg AS	57.00 ± 6.64***	64.0 ± 3.20***	275.01 ± 25.61*	282.71 ± 8.65**
3.75 mg/kg GLZ + 300 mg/kg AS	51.67 ± 3.53**	68.67 ± 3.69***	232.02 ± 1.11*	311.17 ± 5.91***
2.5 mg/kg GLZ + 300 mg/kg AS	50.33 ± 5.51*	50.33 ± 2.90***	242.05 ± 3.57*	290.54 ± 6.83***
1.25 mg/kg GLZ + 300 mg/kg AS	47.67 ± 1.70**	35.0 ± 1.67***	283.30 ± 4.40*	276.30 ± 7.81**
3.75 mg/kg GLZ + 50 mg/kg MET	69.33 ± 2.89***	68.67 ± 2.01***	378.07 ± 1.46***	376.10 ± 5.13***
3.75/2.5/1.25 mg/kg GLZ + 300 mg/kg AS	54.00 ± 4.90**	38.67 ± 3.72***	243.75 ± 5.82*	264.13 ± 16.82**

Each value represents the mean ± SEM for each group ($n = 6$)

GLZ Glipizide, MET metformin, AS *Annona squamosa*, 3.75/2.5/1.25 mg/kg GLZ: reduction in dose of Glipizide every 10th day

* $P < 0.05$ diabetic control vs treated group, ** $P < 0.01$ diabetic control vs treated group, *** $P < 0.001$ diabetic control vs treated group, **** $P < 0.001$ normal vs diabetic control

Table 5 Blood glucose levels of different dose combination of *A. squamosa* and glipizide

Group	Urine output/24 h			
	Initial	10th day of treatment	20th day treatment	30th day treatment
Normal control	13.14 ± 00.57	12.67 ± 0.33	15.01 ± 0.57	13.01 ± 0.57
Diabetic control	54.33 ± 06.89****	63.33 ± 4.41****	68.33 ± 9.71****	69.33 ± 7.18****
5 mg/kg Glipizide	46.10 ± 03.79*	73.33 ± 4.41	45.33 ± 4.28	34.67 ± 1.41***
300 mg/kg AS	66.67 ± 07.26*	78.33 ± 4.41	47.02 ± 6.92	40.09 ± 0.57**
3.75 mg/kg GLZ + 300 mg/kg AS	46.33 ± 06.96*	60.02 ± 7.638	55.67 ± 4.70	43.22 ± 0.57***
2.5 mg/kg GLZ + 300 mg/kg AS	56.67 ± 04.63*	40.00 ± 6.807	38.33 ± 6.11	42.19 ± 0.13***
1.25 mg/kg GLZ + 300 mg/kg AS	53.67 ± 03.84*	32.33 ± 9.838	32.67 ± 7.79	51.01 ± 1.10***
3.75 mg/kg GLZ + 50 mg/kg MET	53.01 ± 13.65*	32.67 ± 9.493	33.33 ± 7.60	31.98 ± 0.57***
3.75/2.5/1.25 mg/kg GLZ + 300 mg/kg AS	40.33 ± 02.90*	32.02 ± 8.50	33.33 ± 7.21	52.08 ± 0.57**

Each value represents the mean ± SEM for each group ($n = 6$)

GLZ Glipizide, MET Metformin, AS *Annona squamosa*, 3.75/2.5/1.25 mg/kg GLZ: reduction in dose of Glipizide every 10th day

* $P < 0.05$ diabetic control vs treated group, ** $P < 0.01$ diabetic control vs treated group, *** $P < 0.001$ diabetic control vs treated group, **** $P < 0.001$ normal vs diabetic control

Forced swim test for depressive behaviour

In depressive behaviour study of forced swimming the diabetic rats showed significantly ($P < 0.001$) reduced time to immobilisation (Table 4). With administration of *A. squamosa* alone, in reduced and reducing dose combination with Glipizide, time of immobilisation was prolonged significantly ($P < 0.01$). After 15 days treatment, no significant results were observed in reduced dose combinations. Better activity was found in reduced and reducing dose combinations after 30 days of treatment.

Urine volume

Analysis of Table 5 shows that there is significant change in urine volume when disease control animals were compared with reduce and reducing dose of *A. squamosa* and with Glipizide.

Discussion

The fundamental mechanism underlying hyperglycemia in diabetes mellitus involves over-production (excessive hepatic

glycogenolysis and gluconeogenesis) and decreased utilisation of glucose by tissues [24]. Although various types of oral hypoglycemic agent are currently available along with insulin to treat diabetes mellitus, there is a growing interest in herbal remedies due to ever more side-effects associated with existing therapeutic hypoglycemic agents [25]. Further, there is a need to develop safer and more effective antidiabetic drugs. Nowadays, researchers are working with new types of combinations of herbal and synthetic drug to treat diseases in order to achieve better success with therapeutic goals as compared to treatment with synthetic drugs alone. In the present study, we evaluated the effect of *Annona squamosa* when given along with different fractions of Glipizide in type-2 diabetes and its complications in rats. The present investigation indicates the better antihyperglycemic response using the reduced and reducing dose combinations of Glipizide and *A. squamosa*. Some of these combinations show protective effects in diabetic complications such as fatty liver, kidney dysfunction, decreased antioxidant defense, elevated lipid levels, and also improved the diabetic encephalopathy in type-2 diabetes.

Blood glucose levels were reduced significantly when animals were treated with Glipizide alone, *A. squamosa* alone and Glipizide in reduced doses and reducing dose combination with *A. squamosa* as compared with the diabetic control group. The results show that combinations *A. squamosa* with 3.75 mg/kg and 2.5 mg/kg doses of Glipizide gives very similar results as Glipizide alone. The possible mechanism of *A. squamosa* hypoglycemic action may be through potentiation of pancreatic secretion of insulin from islet β -cell or due to enhanced transport of blood glucose to the peripheral tissue. This was evidenced clearly by the increased level of insulin in diabetic rats treated with *A. squamosa* [26, 27]. A further chemical constituents of *A. squamosa* responsible for antidiabetic activity is quercetin [18]. Glipizide is a drug that enhances secretion of insulin from pancreatic cells, and increases the peripheral utilisation of insulin. So, in combination, both drugs display a synergistic action with each other, and blood glucose levels were better controlled even in reduced dosage when given with *A. squamosa*. From the results of our study, we conclude that *A. squamosa*, when given in combination with Glipizide, can create the possibilities of dose reduction of Glipizide up to 2.5 mg/kg.

Numerous studies have revealed lowered antioxidant and enhanced peroxidative status in diabetes mellitus [28]. In the current study, activities of SOD and CAT were reduced significantly in the kidneys of diabetic rats. These observations emphasise the critical importance of maintaining the antioxidant potential of the pancreatic β -cells in order to ensure both their survival and insulin secretion capacity during times of increased oxidative stress. The decreased activities of SOD and CAT in kidneys during diabetes

mellitus may be due to the production of reactive oxygen free-radical, which can itself reduce the activity of these enzymes.

Reduced glutathione (GSH) is a potent-free radical scavenger within the islet of β -cells and is an important factor acting against the progressive destruction of β -cells following partial pancreatectomy [29]. Depletion of GSH results in enhanced lipid peroxidation. This can cause increased GSH consumption and can be correlated to the increase in the level of oxidised glutathione (GSSG). Treatment with *A. squamosa* alone, along with reduced and reducing dose combinations of Glipizide results in the elevation of GSH levels in a better way than with Glipizide alone. This protects the cell membrane against oxidative damage by regulating the redox status of protein in the membrane [30]. SOD and CAT destroy peroxides and play a significant role in providing antioxidant defense in many organisms. CAT is involved in the elimination of H_2O_2 . SOD acts to convert the dismutase superoxide radical to H_2O_2 ; the functions of all three enzymes are interconnected and a lowering of their activity results in the accumulation of lipid peroxides and increased oxidative stress in diabetic rats. Treatment with *A. squamosa* in reduced and reducing dose combination of Glipizide increased the activity of these enzymes as compared to treatment with *A. squamosa* and Glipizide alone. Better activity was found in reduced dose 3.75 and 2.5 mg/kg Glipizide combination with *A. squamosa* compared to Glipizide alone.

The increase in the activities of plasma SGPT and SGOT indicated that diabetes mellitus may induce hepatic dysfunction. The enzymes associated directly with the conversion of amino acids to keto acids are SGPT and SGOT, the activities of both being increased in the diabetic condition. An increase in the activities of SGPT and SGOT are also reported in the liver of diabetic animals. Treatment with *A. squamosa* alone or in combination with the Glipizide normalised these enzyme activities. The reduced and reducing dose combinations of Glipizide showed results very similar to Glipizide alone. Similarly, increased activities of SGPT and SGOT in the diabetic liver were also reported [26]. The increased protein catabolism accompanying gluconeogenesis and urea formation that is seen in the diabetic state might be responsible for the elevation of these tissue transaminases. The rise in SGOT activity is due to hepatocellular damage and is usually accompanied by a rise in SGPT. This might be the reason for the elevated activities of these enzymes, which were brought back to near normal values by *A. squamosa* treatment alone and in combination with the Glipizide and *A. squamosa*, thus resulting in better control of blood sugar levels.

Emerging epidemiological data indicates that diabetes is a potential predisposing factor for neuropsychiatric deficits

as stroke, cerebrovascular diseases, diabetic encephalopathy, depression and anxiety. Diabetic encephalopathy, characterised by impaired cognitive functions and neurochemical and structural abnormalities, involves direct neuronal damage caused by intracellular glucose. The direct glucose toxicity in the neurons is due especially to increased intracellular glucose oxidation, which leads to an increase in reactive species production [31]. Oxidative stress seems to play a central role in brain damage both in human and experimentally diabetic rats. Recently, it was reported that oxidative damage to rat synapses contributes to the cognitive deficit [32]. In the present study, STZ-treatment produced a significant increase in blood-glucose level. *A. squamosa* treatment along with Glipizide significantly prevented this rise in blood-glucose level and maintained the body weight of diabetic animals. This effect was observed due to the strong antioxidant and direct stimulatory action of *A. squamosa* treatment along with Glipizide on the pancreatic beta-cell that could contribute towards hypoglycemic activity.

In spontaneously diabetic rats, increased anxiety in the plus maze has been well reported [33]. Time of open arm exploration was lower in diabetic rats, while the treated rats showed an increased time of open arm exploration in the elevated plus maze test. *A. squamosa* treatment along with Glipizide in reduced doses 3.75 and 2.5 mg/kg cured this anxiety behaviour of the animals better than Glipizide alone. In the forced swim test of depressive behaviour, the diabetic animals showed decreased time to immobilisation. When treated with *A. squamosa* along with Glipizide in reduced and reducing doses, the rats showed better results compared to treatment with Glipizide alone. This may lower the chance of further increase in sugar levels due to stress conditions.

Conclusion

The study indicates that, when given in combination Glipizide, *Annona squamosa* not only controls blood glucose levels but also reduces the dosage of Glipizide by 25–50 %. On the basis of the strong outcomes of this study, we suggest that this combination also improves psychopharmacological performance. It was also concluded from the data that *A. squamosa* leaf extract supplementation along with reduced and reducing doses of Glipizide improves the conditions in diabetic complications from lipid peroxidation and antioxidant systems in experimental type-2 diabetic rats. This combination could be useful for prevention not only of diabetes but also of its complications. However, longer duration studies of *A. squamosa* and its isolated compounds with Glipizide on chronic models are necessary to establish long term safety in animals and

humans. This may lead to the development of a potent combination of antidiabetic drugs.

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